

# Chapter 6

## Taylor Expansion

$$f(x) = \sum_{n=0}^{\infty} k_n (x-a)^n$$

$f(z)$  is analytic

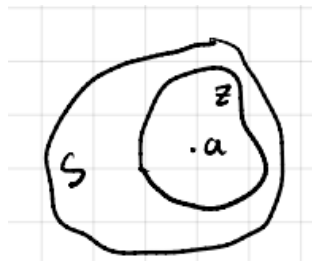
$$\oint_c \frac{f(z)}{z-a} dz = 2\pi i f(a) \quad a \in \Omega$$

$$\oint_c \frac{f(s)}{s-a} ds = 2\pi i f(a)$$

When  $a \rightarrow z$

$$\oint_c \frac{f(s)}{s-z} ds = 2\pi i f(z)$$

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \oint \frac{f(s)}{s-z} ds \\ &= \frac{1}{2\pi i} \oint f(s) \frac{1}{s-z} ds \end{aligned}$$



$$\frac{1}{s-z} = \frac{1}{(s-a) - (z-a)} = \frac{1}{s-a} \sum_{n=0}^{\infty} \left( \frac{z-a}{s-a} \right)^n$$

$$\sum_{n=0}^{\infty} \left( \frac{z-a}{s-a} \right)^n = \frac{\left( \frac{z-a}{s-a} \right)^0 - \left( \frac{z-a}{s-a} \right)^{\infty}}{1 - \frac{z-a}{s-a}} = \frac{s-a}{(s-a) - (z-a)} \left| \frac{z-a}{s-a} \right| < 1$$

$$\begin{aligned}
f(z) &= \frac{1}{2\pi i} \oint_C f(s) \frac{1}{s-z} dz \\
&= \frac{1}{2\pi i} \oint_C f(s) \frac{1}{s-a} \sum_{n=0}^{\infty} \frac{(z-a)^n}{(z-a)^n} ds \\
&= \frac{1}{2\pi i} \oint_C f(s) \sum_{n=0}^{\infty} \frac{1}{(s-a)^{n+1}} \cdot (z-a)^n ds \\
&= \sum_{n=0}^{\infty} \frac{1}{2\pi i} \oint_C f(s) \frac{1}{(s-a)^{n+1}} (z-a)^n ds \\
&= \sum_{n=0}^{\infty} \left[ \frac{1}{2\pi i} \oint_C f(s) \frac{1}{(s-a)^{n+1}} ds \right] (z-a)^n \\
&= \sum_{n=0}^{\infty} C_n (z-a)^n \\
C_n &= \frac{1}{2\pi i} \oint_C f(s) \frac{1}{(s-a)^{n+1}} ds \\
&= \frac{f^{(n)}(a)}{n!}
\end{aligned}$$

## 1. Taylor expansion

$f(z)$  is analytic, or  $f(z)$  is analytic in the neighborhood of point  $a$

$$\begin{aligned}
f(z) &= \sum_{n=0}^{\infty} C_n (z-a)^n \\
C_n &= \frac{f^{(n)}(a)}{n!}
\end{aligned}$$

**Example**



$$f(z) = \sum_{n=0}^{\infty} C_n (z-a)^n = \sum_{n=0}^{\infty} d_n (z-b)^n$$

$$c_n = \frac{f^{(n)}(a)}{n!} \quad d_n = \frac{f^{(n)}(b)}{n!}$$

When  $a = 0$

$$(e^z)^{(n)} = e^z|_{z=0} = 1$$

$$e^2 = \sum_{n=0}^{\infty} \frac{1}{n!} \cdot z^n$$

When  $a = 1$

$$(e^2)^{(n)} = e$$

These are some examples

$$e^2 = \sum_{n=0}^{\infty} \frac{e}{n!} (z-1)^n$$

$$e^3 = \sum_{n=0}^{\infty} \frac{e}{n!} - 2^n \Rightarrow e^2 = \frac{\infty}{n-1}$$

$$e^{\sqrt{2}} = \sum_{n=0}^{\infty} \frac{e}{n!} (\sqrt{2}-1)^n$$

$$e^{\sin\theta+1} = \sum_{n=0}^{\infty} \frac{e}{n!} (\sin\theta)^n$$

$$\frac{1}{1-z^{-1}} = \sum_{n=0}^{\infty} z^{-n} = \frac{z}{z-1}$$

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n \quad |z| < 1$$

$$\frac{1}{1-z^2} = \sum_{n=0}^{\infty} c_n z^n = \sum_{n=0}^{\infty} z^{2n}$$

$$\frac{1}{1+z} = \sum_{n=0}^{\infty} (-z)^n = \sum_{n=0}^{\infty} (-1)^n \cdot z^n$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$e^z = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$$

$$e^{iz} = \sum_{n=0}^{\infty} \frac{1}{n!} (iz)^n = \sum_{n=0}^{\infty} \frac{1}{n!} \cdot i^n z^n$$

$$e^{-iz} = \sum_{n=0}^{\infty} \frac{1}{n!} (-iz)^n = \sum_{n=0}^{\infty} \frac{1}{n!} \cdot (-i)^n z^n$$

$$\cos z = \frac{1}{2} [e^{iz} + e^{-iz}]$$

Left side:

$$1 - i + \frac{1}{2}(-1) - \frac{1}{3!}(-i) + \frac{1}{4!}$$

Right side :

$$1 + i - \frac{1}{2}(-1) - \frac{1}{3!}i + \frac{1}{4!}$$

We can see that

$$\cos z = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n}$$